# SIMATS ENGINEERING

## SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

### Department of Computer Science and Engineering

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# ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING

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| **Course Code:** | DSA0502 | **Course Title:** | Query Processing for Data Science |
| **CO Assessed:** | CO5 | **Assessment:** | Tool 2– **INDUSTRY PROBLEM SOLVING** |
| **Weightage:** | 35% | **Total Marks:** | |
| **CO5 Statement:** Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication | | | |
| **Student Name:** | | **Reg. No.:** | |
| **Section / Batch:** | | **Date of Submission:** | |
| **Faculty In-charge:** | **Dr. B.Shyamala Gowri** | **Project Mode:** | Individual Submission |

**Code of Conduct**

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: __________________________

# 1. Manufacturing Production Analysis

# Aim

To import and explore daily manufacturing data (production, machine operating hours, temperature, energy consumption, and defect count) using Pandas, and to develop visualizations that reveal production trends over time, relationships between machine parameters and defects, and the underlying distribution of production values. A scatter matrix, lag plot, and autocorrelation plot are used to uncover hidden production patterns, leading to recommendations for improving manufacturing efficiency.

# Algorithm

1. Import pandas, numpy, and matplotlib.pyplot; import scatter_matrix, lag_plot, and autocorrelation_plot from pandas.plotting.

2. Load the manufacturing dataset into a DataFrame with 'date' as a datetime index, and inspect it using head(), info(), and describe().

3. Plot daily production against date as a line chart to visualize the production trend over time.

4. Create scatter plots of machine operating hours vs. defect count and temperature vs. defect count to study how these parameters relate to defects.

5. Plot a histogram of the production column to study its distribution (spread, skew, and outliers).

6. Generate a scatter matrix of production, machine_hours, temperature, energy_consumption, and defect_count to visualize pairwise relationships and individual distributions at a glance.

7. Generate a lag plot of production (value at t vs. value at t-1) to check for randomness or autocorrelation in the series.

8. Generate an autocorrelation plot of production across multiple lags to identify any repeating/cyclical pattern.

9. Compute the correlation matrix (df.corr()) to quantify the strength of relationships between variables.

10. Interpret all visualizations together and derive efficiency recommendations.

# Output

## Production Trend Over Time

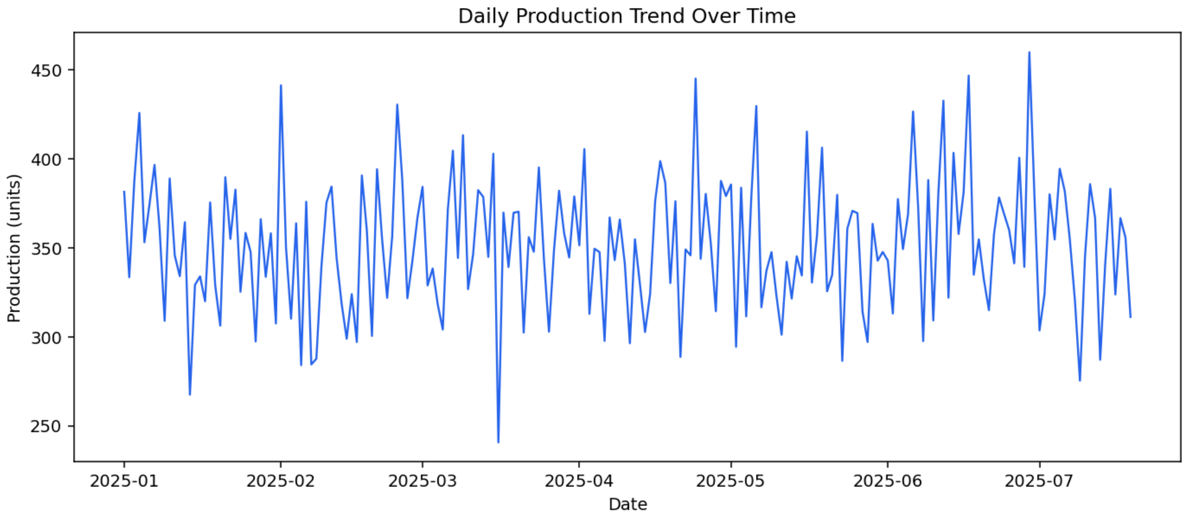


*Fig. 1 — Daily production shows short-term fluctuation around a fairly stable mean, with no strong long-term drift.*

# Machine Parameters vs. Defects

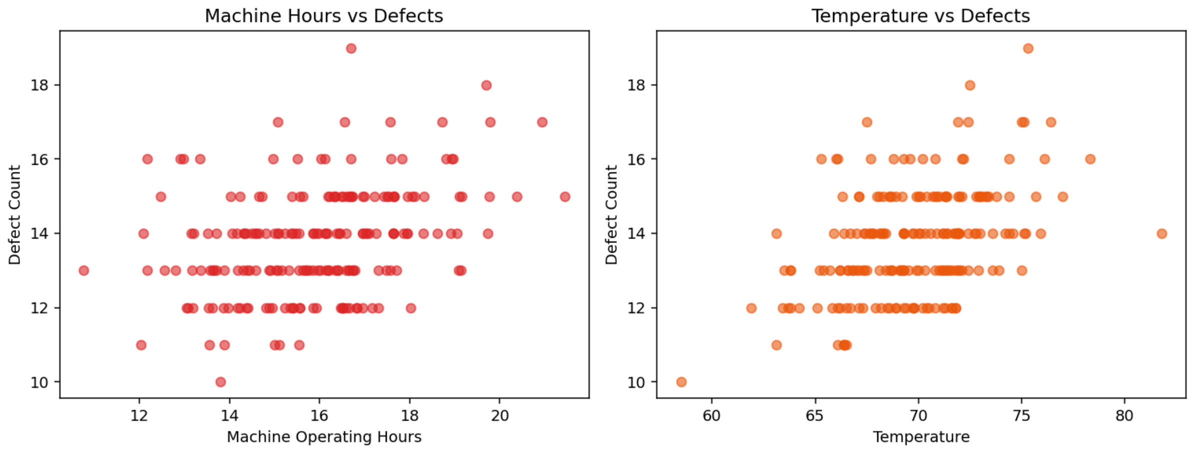


*Fig. 2 — Defect count rises noticeably once machine operating hours and temperature exceed typical operating ranges.*

# Distribution of Production

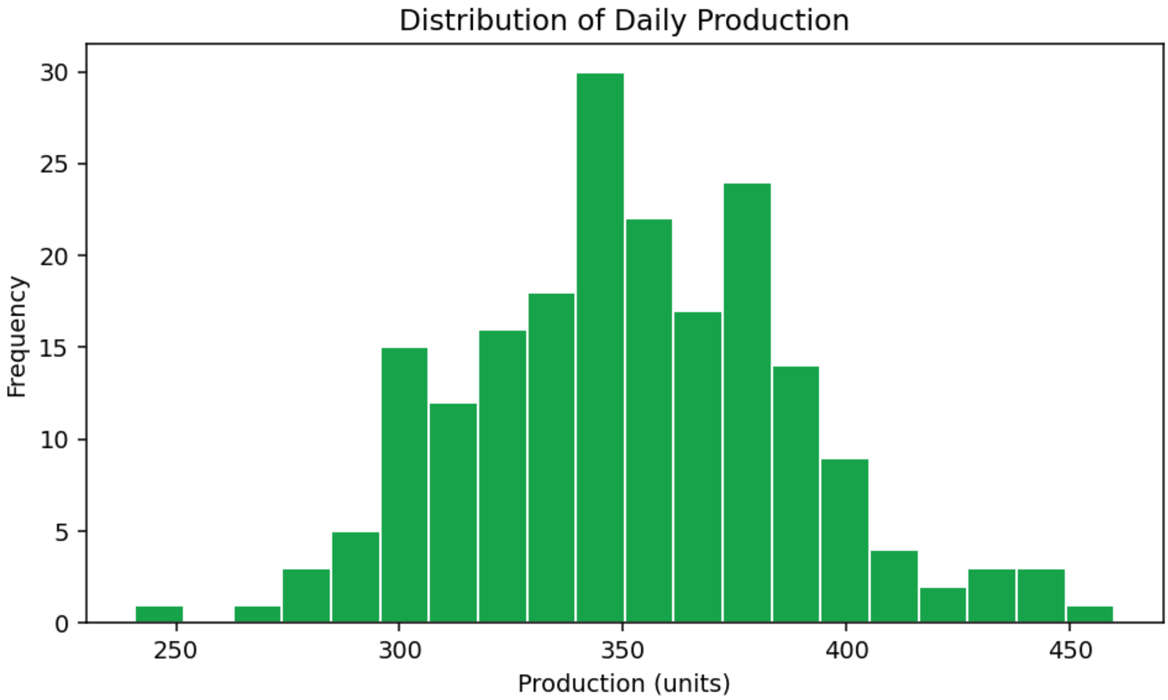


*Fig. 3 — Production values are approximately normally distributed with a slight right skew.*

## Scatter Matrix of Production Parameters

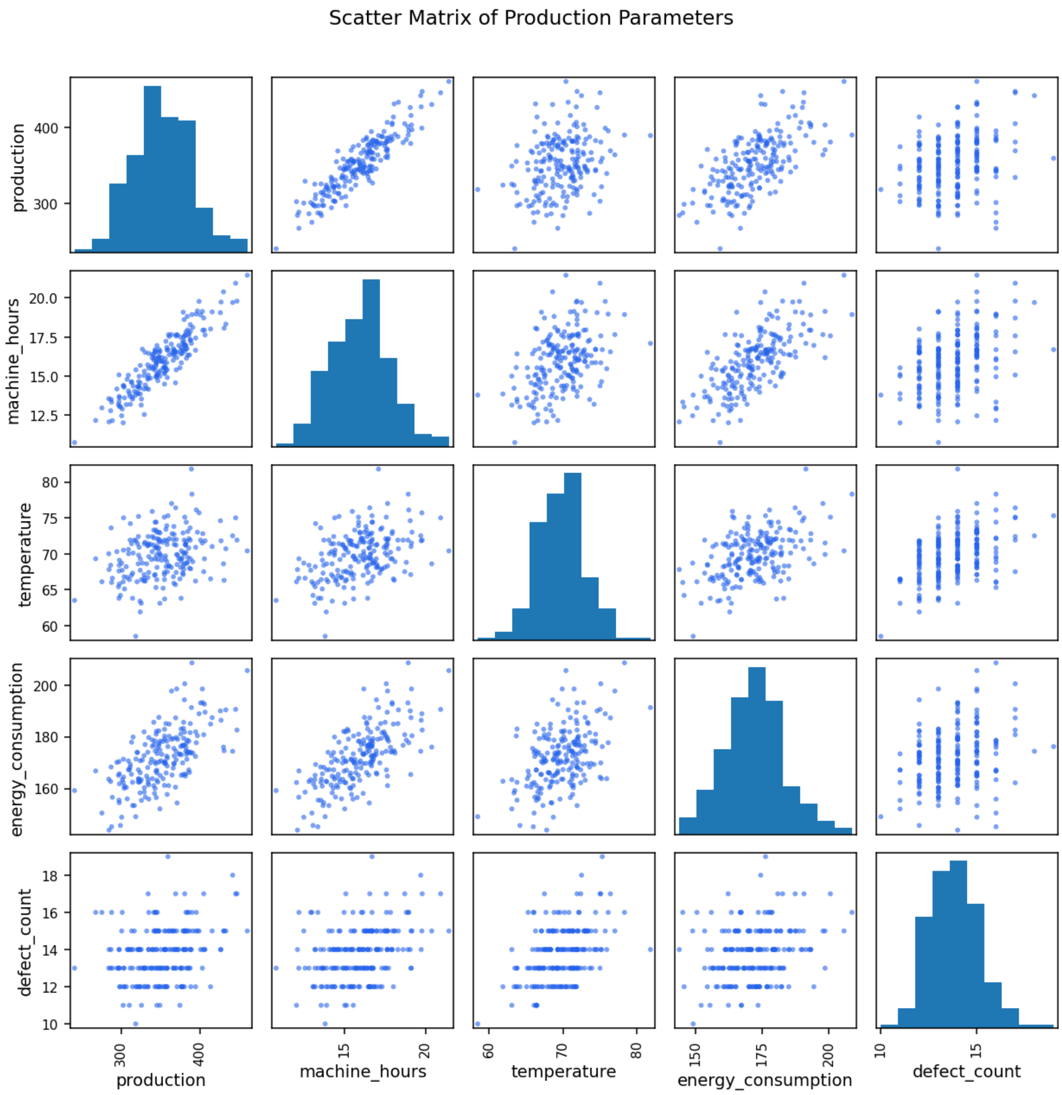


*Fig. 4 — Pairwise relationships show production is strongly, positively related to machine hours, while defects rise with both temperature and machine hours.*

## Lag Plot of Daily Production

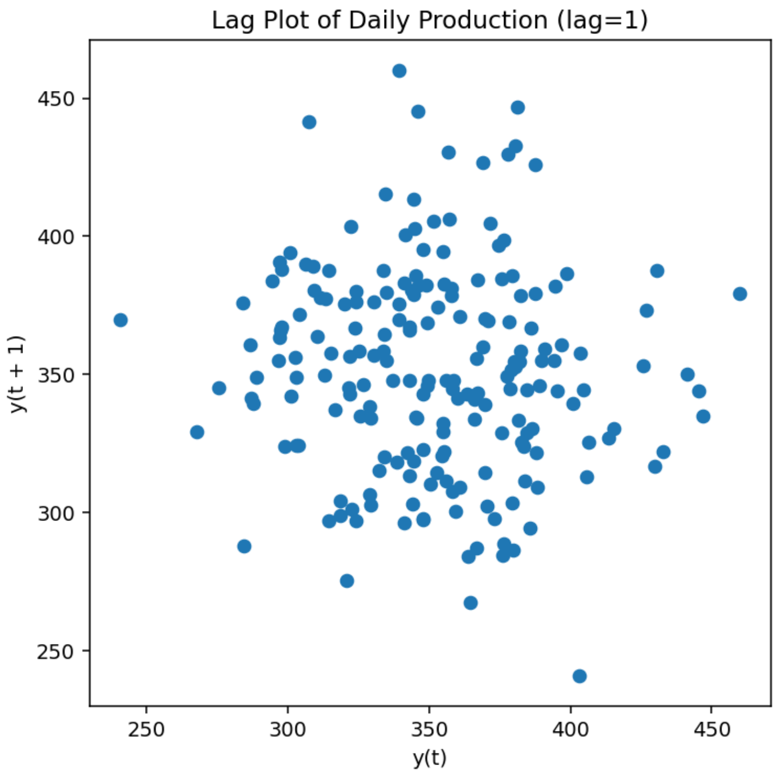


*Fig. 5 — The scattered, non-linear cloud indicates production on a given day is largely independent of the previous day's output (little day-to-day carryover).*

## Autocorrelation Plot of Daily Production

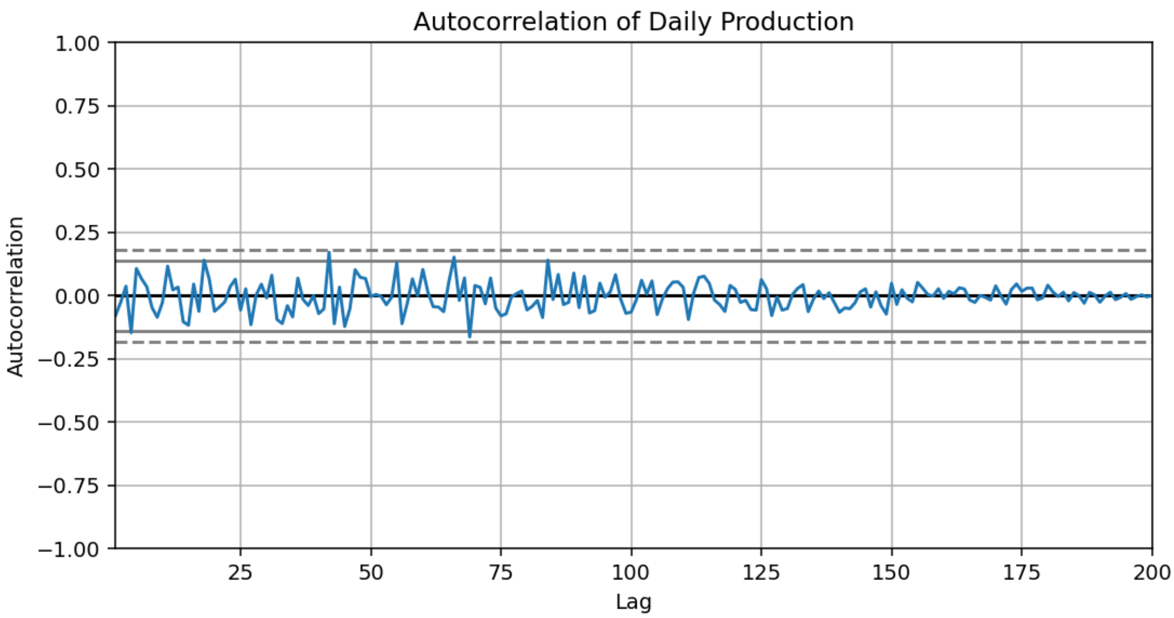


*Fig. 6 — Autocorrelation values stay within the confidence band at nearly all lags, confirming production behaves close to a stationary, random process rather than following a strong cyclical pattern.*

# Observations & Recommendations

- Production correlates strongly with machine operating hours ($r \approx 0.91$) — the single biggest lever for output.
- Defect count increases with both temperature and machine hours, suggesting overheating/over-utilization drives quality issues — consider cooling intervals or preventive maintenance once machines exceed ~18 hours/day.
- Energy consumption tracks machine hours and temperature closely; running machines cooler could reduce energy cost without hurting output much, since energy correlates only moderately with production.
- The lag and autocorrelation plots show no strong day-to-day dependency, so short-term production dips are likely due to operational noise (e.g., staffing, input quality) rather than a systemic cyclical issue — investigate daily operational logs for outlier days.
- Recommend installing temperature monitoring alerts and scheduling maintenance windows when machine hours approach the observed defect-inflection point (~18 hrs/day).

# 2. E-Commerce Sales Pattern Analysis

# Aim

To analyze historical e-commerce data (date, product category, quantity sold, revenue, discount, and customer rating) using Pandas, study how sales evolve over time, identify relationships between discount, quantity sold, and revenue, and use bar charts, line charts, distribution plots, frequency plots, and a scatter matrix to uncover important business patterns.

# Algorithm

1. Import pandas, numpy, and matplotlib.pyplot; import scatter_matrix from pandas.plotting.

2. Load the e-commerce dataset into a DataFrame with columns date, category, quantity_sold, revenue, discount, and customer_rating; inspect with head() and describe().

3. Group the data by date and plot total daily revenue as a line chart to observe the sales trend over time.

4. Group the data by category and plot total revenue as a bar chart to compare category-wise performance.

5. Create scatter plots of discount vs. quantity_sold and discount vs. revenue to examine how discounting affects demand and earnings.

6. Plot a histogram of customer_rating to study the distribution of customer satisfaction.

7. Plot a bar chart of order-record frequency per category to see which categories generate the most transactions.

8. Generate a scatter matrix of quantity_sold, revenue, discount, and customer_rating to visualize pairwise relationships at once.

9. Compute the correlation matrix to quantify how strongly discount relates to quantity and revenue.

10. Interpret the visualizations together to identify actionable business patterns.

# Output

## Revenue Trend Over Time

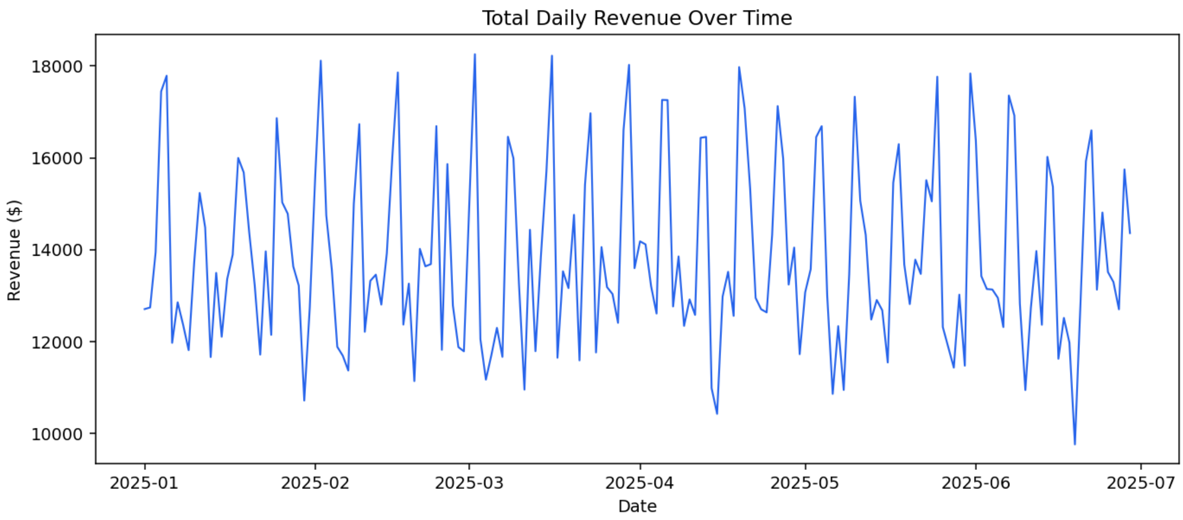


*Fig. 7 — Daily revenue fluctuates with a recurring weekly pattern, spiking on weekends.*

## Revenue by Product Category

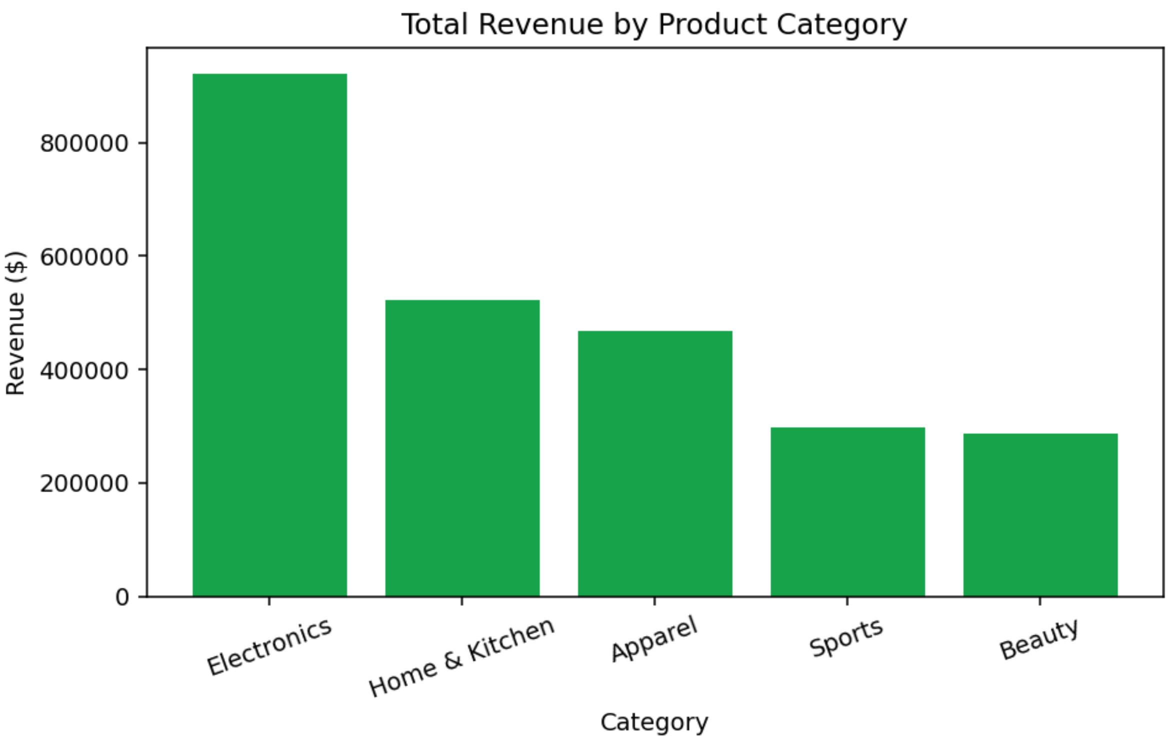


*Fig. 8 — Electronics generates the highest total revenue despite not having the highest unit volume, reflecting its higher price point.*

## Discount vs. Quantity Sold and Revenue

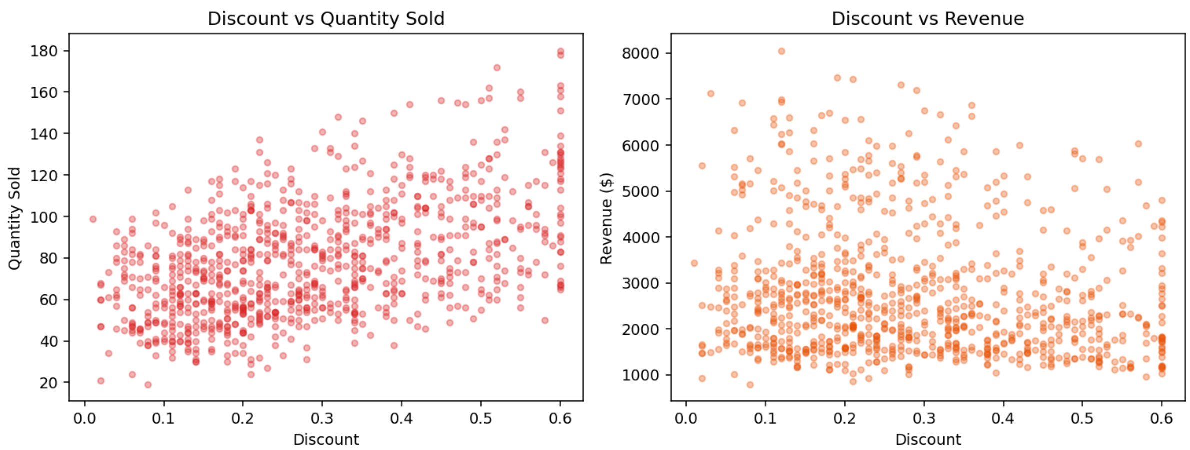


*Fig. 9 — Higher discounts are associated with higher quantity sold but a slight decline in revenue, indicating discounting boosts volume faster than it can offset price reduction.*

# Distribution of Customer Ratings

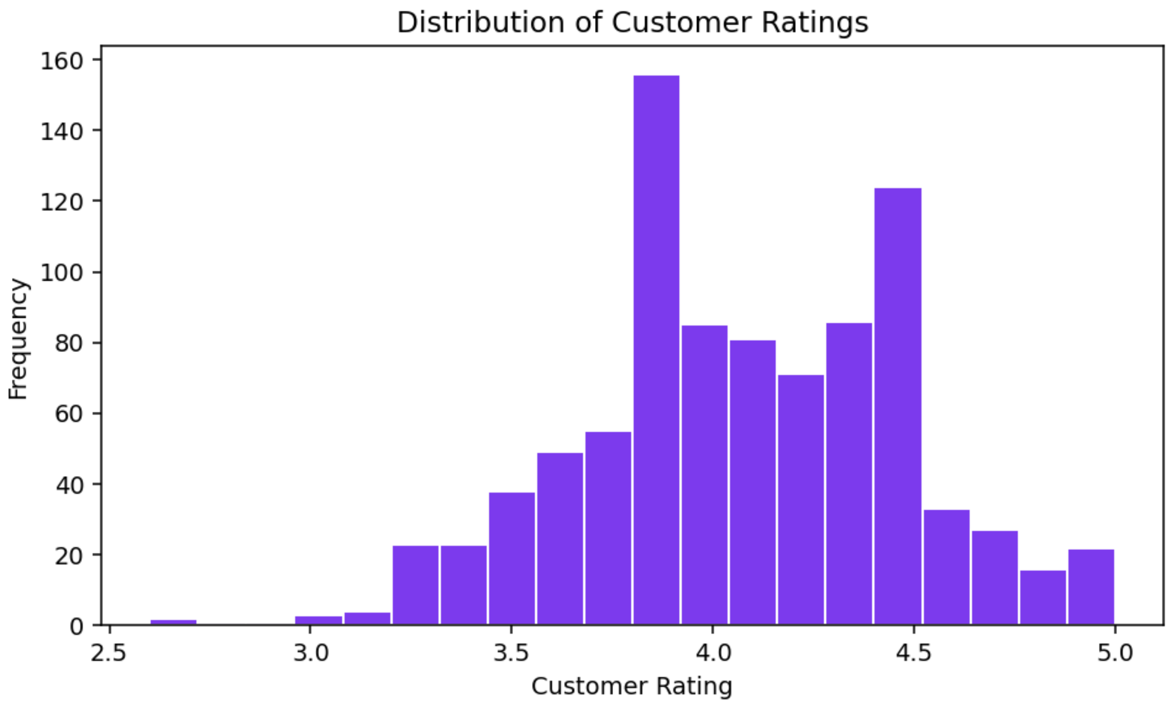


*Fig. 10 — Customer ratings cluster between 3.8 and 4.5, showing generally positive but not extreme sentiment.*

# Order Frequency by Category

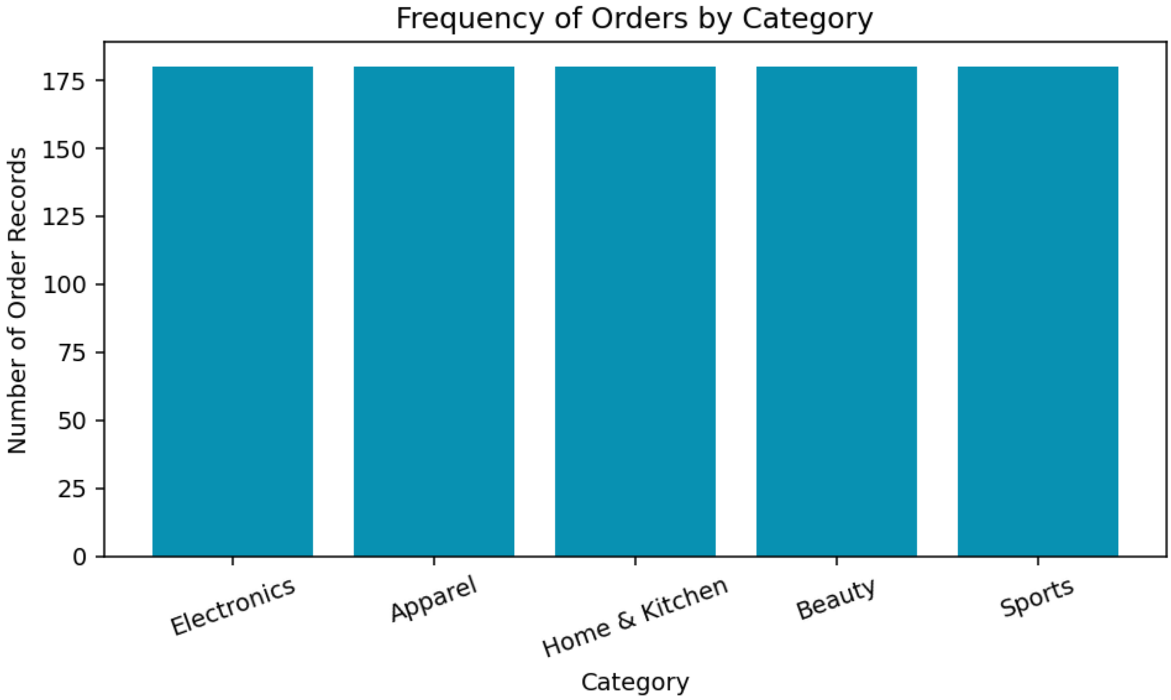


*Fig. 11 — All categories have a similar number of order records since each is sold daily, so revenue differences stem from price and quantity, not listing frequency.*

## Scatter Matrix of Sales Variables

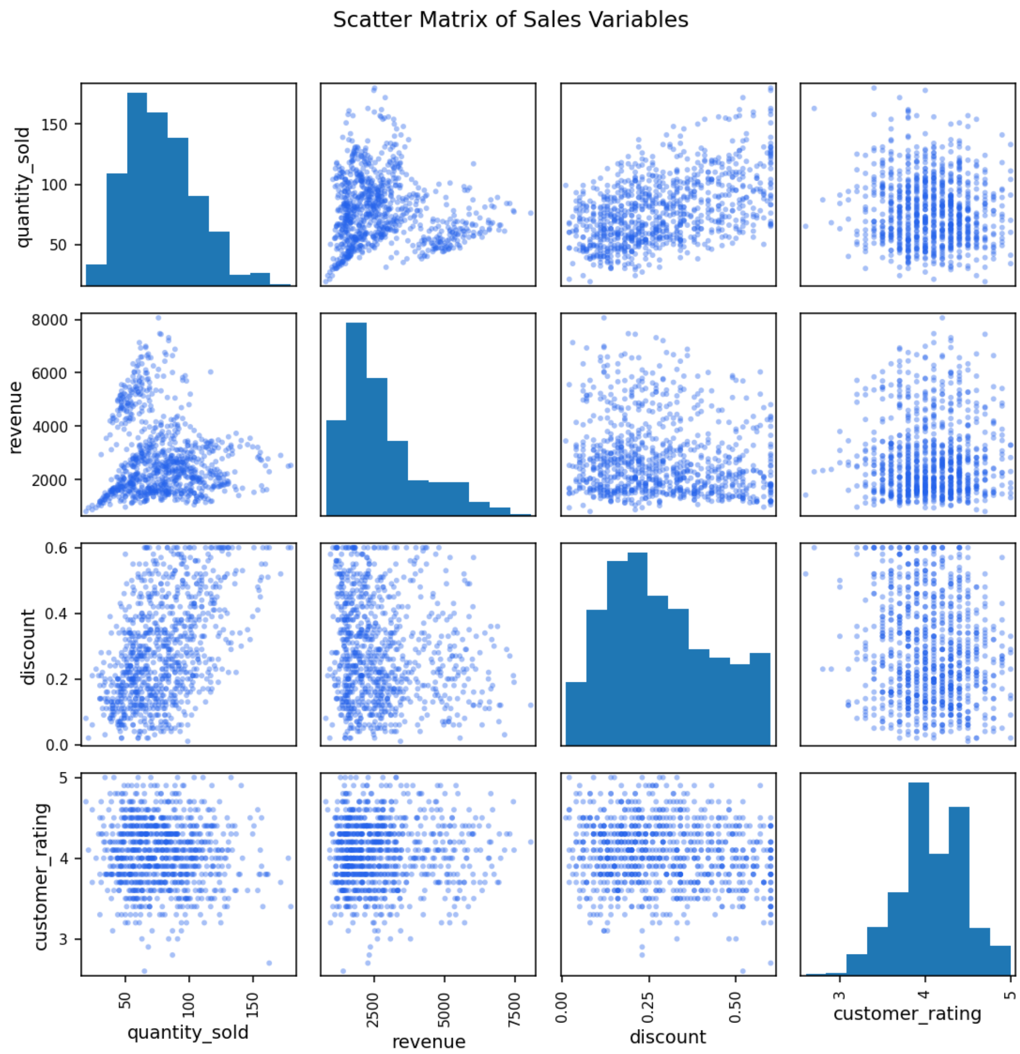


*Fig. 12 — The scatter matrix confirms a positive discount–quantity relationship and a weak negative discount–revenue relationship, while customer rating is largely independent of the other variables.*

# Observations & Recommendations

- Discounting reliably increases quantity sold (r ≈ 0.49) but slightly erodes total revenue (r ≈ -0.14) — deep discounts should be reserved for clearing slow-moving stock rather than applied broadly.
- Electronics is the top revenue driver despite moderate order volume; prioritizing inventory and marketing spend here offers the best return.
- Weekend revenue spikes suggest scheduling promotions and ad spend around weekends to capitalize on naturally higher demand.
- Customer ratings dip slightly as discounts rise, hinting that heavily discounted items may carry lower perceived quality or rushed fulfillment — monitor fulfillment SLAs on high-discount campaigns.
- Beauty and Sports categories have high order frequency but lower revenue contribution — consider bundling or upselling to raise average order value in these categories.